# Pooja Sri

## B.Tech - CSE - (AI&ML)

WOXSEN UNIVERSITY

## Contact


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## Academics

**Woxsen University |** Hyderabad
*2022 - 2026* |**Bachelor of Technology - 3.4/4.0**

**West Berry School| Guntur** *2020 - 2022*
|Intermediate **- 95%**

**West Berry School | Guntur**
*2009 - 2020* | Schooling **- 83%**

## Skills

- Programming Languages: Python, Java
- Machine Learning & Deep Learning
- Computer Vision
- Database Management
- Neural Networks
- Data Structures & Algorithms
- Natural Language Processing (NLP)
- Scripting Languages: HTML, CSS

## Tools

- SQL & MongoDB
- Machine Learning Libraries: TensorFlow, PyTorch, Scikit-Learn, Keras, NLTK
- Data Processing: NumPy, Pandas, Matplotlib
- Web Automation & Scraping: Selenium, BeautifulSoup
- Development Tools: VS Code, Jupyter Notebook
- Frontend&UI/UX:HTML, CSS, JavaScript

## Languages

English - Fluent
Telugu - Fluent
Hindi - Fluent

## Professional Skills

- Problem-Solving & Critical Thinking
- Flexible & Adaptive to Change
- Planning & Coordination
- Teamwork & Collaboration
- Analytical & Strategic Thinking
- Effective Communication & Presentation
- Time Management & Multitasking
- Attention to Detail

## Objective

Proactive, resourceful professional, Highly flexible and adaptive to change, with strong problem-solving and critical-thinking abilities. Strong communicator and team player, fostering collaboration and strategic decision-making to drive success. Detail-oriented and efficient in time management, balancing multiple responsibilities with precision. Committed to continuous learning and innovation, leveraging analytical and strategic thinking to surpass goals and deliver impactful solutions.

## Internship

**InternshipatNIRD-National Institute of Rural Development , Hyderabad**

*May 2025 - June 2025 | NIRD*

- Built a road condition monitoring project using drone and satellite data for rural infrastructure planning.
- Learned QGIS tools for georeferencing, raster analysis, and polygon mapping; gained insights into how satellite data is processed.
- Extracted road segments from GeoTIFFs using Python and OpenCV, without external tools.
- Integrated pre-trained models to assess road quality in percentiles and built a dashboard for automated analysis from drone uploads.
- Contributed to a scalable tool supporting data-driven rural development decisions.

## Projects

**Project - 1 - Self-Supervised Instance Segmentation of Nuclei & Glands- September - November 2025**

- Built a DINOv3 + U-Net self-supervised segmentation model for nuclei and gland detection in H&E histopathology images.
- Leveraged DINOv3 transformer features to boost segmentation on annotated data.
- Achieved strong performance with the DINOv3 + U-Net model (Dice 0.8607, IoU 0.7576) compared to a standalone U-Net (Dice 0.8123, IoU 0.7021).
- Used Grad-CAM to interpret model attention regions for reliable pathology insights.

**Project - 2 - WattWise – Real-Time Fault Detection & Forecasting in Solar PV Systems - August - October 2025**

- Built an ML-based forecasting system using SCADA + weather data to predict solar energy output across two PV plants.
- Trained SARIMAX, LSTM, Random Forest, Prophet, and XGBoost models — with XGBoost achieving 99% accuracy, while others performed below 80%.
- Implemented anomaly detection using residual thresholds to identify potential system faults.
- Developed dashboards for energy forecasts, anomaly alerts, and historical trend visualization to support predictive maintenance.

**Project - 3 - LLM Evaluation & Response Scoring System - January - April 2025**

- Built a dashboard to evaluate chatbot responses across summarization, question-answering, and sentiment analysis tasks using pre-trained LLMs like LLaMA-2 7B, Flan-T5, RoBERTa-large.
- Created a custom dataset tailored to each task for realistic evaluation scenarios.
- Integrated prompt routing to dynamically select the most suitable model based on query type.
- Designed the system in a modular Python framework with custom visualizations for comparative analysis.

**Project - 4 - Bottle fill level detection - August - November 2024**

- Developed areal-time bottlefill leveldetectionsystem usingCNN and computer vision, achieving 91% accuracy in detecting liquid levels inside tinted bottles.
- Optimized dataset quality with lighting adjustments, image capture angles, and format conversions for better performance.
- Used OpenCV for preprocessing and trained a CNN model for accurate and adaptable fill level classification.
- Enhanced system efficiency for real-time analysis and quality control in the alcohol beverage industry.

## Certifications

- March2023 –Website Development, NxtWave
- April 2023 – Data Structures, Coursera
- April 2023 – Core Java, Coursera
- November 2023 – June 2024 – Python, NxtWave
- March 2024 – Introduction to TensorFlow for AI, ML & Deep Learning, Coursera
- March 2024 – Introduction to Deep Learning & Neural Networks with Keras, Coursera